A PROJECT TITLE

ON

Box-Office Revenue Prediction Using Elastic Net Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSIT

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1. Abstract:

The **Movie Analytics and Box Office Revenue Prediction System** is an advanced software application that integrates **Data Structures and Algorithms (DSA)** with **Machine Learning** to provide efficient movie search, analytics, recommendation, and revenue prediction. The project is designed to demonstrate the practical application of advanced algorithmic techniques such as **String Algorithms, Dynamic Programming, Network Flow, Approximation Algorithms, and Randomized Algorithms**, while also incorporating **Elastic Net Regression** for predictive analysis.

The system maintains a large movie dataset containing information such as movie title, genre, budget, runtime, cast, director, ratings, and release details. Users can efficiently search for movies using **Knuth-Morris-Pratt (KMP)** and **Rabin-Karp** string matching algorithms. Misspelled queries are corrected using **Edit Distance** implemented through **Dynamic Programming**, while **Suffix Arrays** are utilized to identify similar movie titles and improve search accuracy.

To analyze relationships among actors, directors, genres, and movies, the system models these entities as graphs. **Network Flow algorithms**, including **Ford-Fulkerson** and **Edmonds-Karp**, are applied to solve recommendation, assignment, and influence analysis problems. Approximation and randomized techniques are incorporated to optimize search performance and recommendation quality for large datasets.

The predictive component of the project employs **Elastic Net Regression** to estimate box-office revenue using features such as production budget, genre, runtime, cast popularity, director popularity, IMDb rating, and release season. Elastic Net combines L1 and L2 regularization to reduce overfitting and improve prediction accuracy while handling correlated features effectively.

The proposed system demonstrates how advanced data structures, algorithmic paradigms, and predictive analytics can be integrated into a single intelligent platform. It provides fast search, optimized recommendations, efficient data processing, and accurate revenue prediction, making it a comprehensive academic project that showcases both advanced algorithm design and practical machine learning applications.

2. INTRODUCTION:

The entertainment industry generates enormous amounts of movie-related data, making efficient search, analysis, and revenue prediction increasingly important. Traditional movie search systems primarily rely on database queries and basic filtering techniques, which often become inefficient when processing large datasets or handling misspelled search queries. Furthermore, predicting a movie's commercial success requires analysing multiple factors simultaneously.

This project proposes a **Movie Analytics and Box Office Revenue Prediction System** that combines advanced algorithmic techniques with machine learning. String matching algorithms such as **KMP** and **Rabin-Karp** provide efficient movie search, while **Dynamic Programming** enables typo correction through Edit Distance computation. **Network Flow algorithms** analyze relationships among movies, actors, and directors to support intelligent recommendations. Additionally, **Elastic Net Regression** predicts box-office revenue based on important movie characteristics.

The project demonstrates how advanced data structures and algorithms can solve real-world computational problems while improving system efficiency, scalability, and prediction accuracy.

3. SOFTWARE REQUIREMENTS:

Programming Language
- Java (DSA Implementation)
- Python (Elastic Net Regression Model)

IDE
- IntelliJ IDEA / Eclipse (Java)
- Visual Studio Code / Jupyter Notebook (Python)

Libraries

Java
- File Handling
- Swing/JavaFX (Optional GUI)

Python
- NumPy
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn

Database
- CSV Dataset / MySQL (Optional)

Operating System
- Windows 10/11
- Linux
- macOS

Dataset
- TMDB Movie Dataset
- IMDb Movie Dataset
- Kaggle Movie Revenue Dataset

Hardware Requirements
- Intel Core i5 or above
- 8 GB RAM (minimum)
- 10 GB Free Storage